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MORNING

04 DEC 2023

[Total No. of Questions: 09]

[Total No. of Pages: 2]

Uni. Roll No.

Program: B.Tech. (Batch 2018 onward)

Semester: 3

Name of Subject: Data Structures

Subject Code: PCIT-101

Paper ID: 16040

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) List the different types of queues and their key characteristics.
- b) List the different representations of graphs.
- c) List the basic operations that can be performed on common data structures.
- d) Describe the concept of collision resolution in hashing.
- e) Differentiate between time complexity and space complexity.
- f) Name two operations that can be performed on a doubly linked list.

Part – B

[Marks: 04 each]

- Q2.** Develop an algorithm to sort a list of numbers.
- Q3.** Compare and contrast the sequential and linked representations of trees, highlighting the advantages and disadvantages of each.
- Q4.** Explain the operations that can be performed on linear and multi-dimensional arrays.
- Q5.** Devise a method for balancing an AVL tree after a series of insertions and deletions.
- Q6.** Develop a recursive solution for the Tower of Hanoi problem. Explain the underlying principles of your solution.
- Q7.** Evaluate the advantages and disadvantages of using a linear queue versus a circular queue.

Part – C

[Marks: 12 each]

- Q8.** Explain the key terminology associated with trees, including concepts like root, leaf, parent, and child nodes.

OR

Explain the concept of Big O notation and its significance in algorithm analysis. Evaluate the trade-offs between time complexity and space complexity in algorithm design.

- Q9.** Implement the insertion operation for a linear linked list. Compare the advantages and disadvantages of linear linked lists and doubly linked lists. Apply the bubble sort algorithm to sort a given list of numbers.

OR

Evaluate the applications of stacks, specifically in the context of converting infix to prefix and postfix expressions. How does the use of stacks simplify these operations?
